getting rid of the residual malignant cells after the gross tumour has been removed. The best results in the treatment of cancer is to follow an integrated approach to therapy, combining surgery, radiotherapy, chemotherapy and immunotherapy.

7.6 Adolescence – drug and alcohol abuse

Adolescence begins with a period of rapid physical and sexual development called **puberty** to maturity at 12 to 19 years of age. Adolescence is also a highly dynamic period of psychological and social changes in individuals. Adolescents are vulnerable to group (peer) pressure and many youngsters are pushed into experimenting with drugs and alcohol. Proper education and guidance would enable youth to say no to drugs and alcohol and to follow a healthy life style.

Alcohol is a psychoactive drug, which acts on the brain, affecting a person's mind and behaviour. It is a depressant, which slows down the activity of the nervous system. The intake of certain drugs for a purpose other than their normal clinical use in an amount and frequency that impair one's physical, physiological and psychological functions is called **drug abuse**.

The drugs which are commonly abused include **opioids, cannabinoids, coca-alkaloids, barbiturates, amphetamines and LSD**.

Opioids are drugs which bind to specific opioid receptors present in the **central nervous system** and **intestinal tract**. Heroin (smack) is chemically diacetyl morphine, which is white, odourless and bitter crystalline compound. It is obtained by acetylation of morphine, which is extracted from flowers of the poppy plant **(Fig. 7.19)**. **Morphine** is one of the strongest pain killer and is used during surgery. It is the most widely abused narcotic drug which acts as a depressant and slows down body functions.



Fig. 7.19 Opium plant (poppy plant)

Cannabinoids are a group of chemicals obtained from *Cannabis sativa*, the Indian **hemp plant (Fig. 7.20).** Natural cannabinoids are the main source of **marijuana, ganja, hashish** and **charas.** It interferes in the transport of the neurotransmitter, dopamine and has a potent stimulating action on the CNS, producing increased energy and a sense of euphoria.

Table 7.9 Classification of drugs

Group	Drugs	Effects
Stimulants	Amphetamines, cocaine, nicotine and tobacco	Accelerates the activity of the brain
Depressants	Alcohol, Barbiturates, Tranquilizers	Slows down the activity of the brain
Narcotic/ Analgesics	Opium, Morphine	Act as depressants on the Central Nervous System
Hallucinogens	Lysergic acid diethylamide (LSD), Phencyclidine	Distorts the way one sees, hears and feels
Stimulants, Depressants, Hallucinogens	Bhang (Marijuana), Ganja, Charas	Stimulating action on the CNS and affects the cardiovascular system

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